



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd u/s 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043
Phone No. 7815926816, www.klh.edu.in

OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026-27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 10

Team No.: 03

Project Title: Linux System Information and Resource Monitoring Tool

Team Members

Roll Number	Student Name	Signature
2520090090	Ch. Sai Vihamsh	
2520090016	P.Rahul	
2520090008	B. Sathish kumar	

1. Abstract

The Linux System Information and Resource Monitoring Tool is an Operating Systems project designed to monitor and display important information about a Linux-based computer system. In a computer system, it is important to understand how system resources such as CPU, memory, disk space, and running processes are being utilized. Manually checking these resources can be difficult and time-consuming for users.

The proposed project develops a simple Linux-based monitoring tool that collects and displays real-time system information using Linux system commands, shell scripting, and system files. The tool will provide information such as CPU utilization, memory usage, disk usage, system uptime, running processes, and basic system details. It will also help users identify processes that consume high CPU or memory resources.

The project demonstrates important Operating Systems and System Programming concepts, including process management, CPU and memory management, file systems, resource utilization, system calls, and interaction with the Linux /proc file system. The tool will provide a simple and user-friendly interface through which users can easily observe the current state of the system.

The expected outcome is a lightweight Linux monitoring application that helps users understand system resource usage while providing practical knowledge of how an Operating System manages and provides information about hardware and software resources.

2. Problem Statement

In a Linux computer system, various resources such as CPU, memory, disk space, and running processes are continuously used by applications and system services. Users may find it difficult to understand how these resources are being utilized and which processes are consuming more system resources. Existing system monitoring utilities may also provide large amounts of information that can be difficult for beginners to understand.

Therefore, there is a need for a simple Linux-based System Information and Resource Monitoring Tool that collects important system information and presents it in an easy-to-understand format. The tool will monitor CPU usage, memory usage, disk usage, system uptime, and running processes, helping users observe the current state of the system and identify resource-intensive processes.

This project is relevant to Operating Systems and Systems Programming because it demonstrates how an operating system manages hardware and software resources. It involves concepts such as process management, CPU scheduling, memory management, file systems, system calls, and the Linux /proc file system. By developing this tool, the project provides practical understanding of how system-level information can be obtained from Linux and how operating system resources are monitored and managed.

3. Objectives

1. To develop a Linux-based tool for collecting and displaying important system information such as CPU, memory, disk usage, and system uptime.
2. To monitor running processes and identify processes that consume high CPU and memory resources.
3. To demonstrate Operating System concepts such as process management, memory management, file systems, system calls, and resource utilization.
4. To provide a simple and user-friendly monitoring interface that helps users understand the current status and resource usage of a Linux system.

4. Proposed Methodology

The project will be designed and implemented as a Linux-based system monitoring tool using shell scripting and Linux system utilities. The tool will collect system information from Linux commands and system files, particularly the /proc file system, and process the collected data to display useful resource information.

The main functionality will include monitoring CPU usage, memory usage, disk usage, system uptime, and running processes. The tool will periodically collect the required information and display it in a simple and understandable format. For process monitoring, the tool will identify processes using high CPU or memory resources.

The basic flow of the system will be: start the tool → collect system information → read resource and process details → calculate or process usage values → display the information → continuously refresh the data. This approach will demonstrate important Operating System concepts such as process management, memory management, file systems, CPU utilization, and system-level resource monitoring.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
Process Management	To monitor running processes and display information such as process ID, CPU usage, memory usage, and process state.
Memory Management	To monitor RAM usage and display total, used, and available memory of the Linux system.
File Systems and /proc File System	To obtain system and process information from Linux virtual files such as /proc/cpuinfo, /proc/meminfo, and /proc/[PID]/status.
Linux System Calls	To interact with the operating system and obtain system-level information and resource details.
Shell as a User-Space Program	To execute Linux commands and combine system utilities for collecting and displaying monitoring information.

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520090008	Sathish Kumar	Project design, system information module, CPU and memory monitoring, and overall integration
2520090090	Sai Vihamsh	Process monitoring module, process information collection, and high resource-consuming process identification
2520090016	P. Rahul	Disk monitoring module, /proc file system data collection, testing, and documentation

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Operating system platform for developing and running the monitoring tool.
C / C++	Programming language used to implement the system monitoring functionality.
GCC (GNU Compiler Collection)	To compile and build the C/C++ source code in Linux.
Linux /proc File System	To obtain CPU, memory, process, and other system information directly from the Linux kernel interface.

8. Expected Outcome

The project will produce a Linux-based System Information and Resource Monitoring Tool that displays CPU usage, memory usage, disk usage, system uptime, and running processes. It will provide updated resource information through a simple command-line interface and help identify processes consuming high CPU or memory. The project will demonstrate OS concepts such as process management, memory management, /proc file system, system calls, and Linux resource management.

9. GitHub Repository

GitHub Repository Link: <https://github.com/palavalasarahul-blip/OSSP-Project>

Repository Created: yes

☒ Yes ☐ No

Faculty Name and Signature

Faculty Name: Mr. Raghupathi Manthena

Signature: 

Remarks: _____

Faculty Use

Project Approved: approved

☒ Yes ☐ No

Date: 26/11/2021

Faculty Signature: 